

# Companion to Bird & de Moor, Algebra of Programming

## §8. Thinning Algorithms

### §8.1. Thinning

(8.1)

For  $Q : A \rightarrow A$ ,  $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : EA \rightarrow EA$  (8.1).

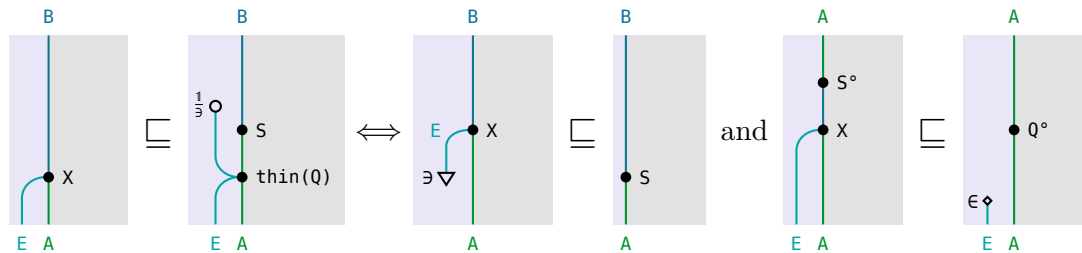
$ys \text{ thin}(Q) \ xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$  the same  $^\circ$  as  $\triangleq \text{est}$ :  $\epsilon X$  runs  $a \rightarrow ys \rightarrow xs$ , member of  $ys$  first, and  $Q^\circ \epsilon$  runs  $a \rightarrow b \rightarrow xs$ , so  $(a, b) \in Q^\circ$  reads  $b \ Q \ a$ ; at  $Q^\circ \leq$  every  $a \in ys$  keeps some  $b \leq a$  in  $xs$ , the end  $\text{est}(\leq)$  picks

(8.1a)

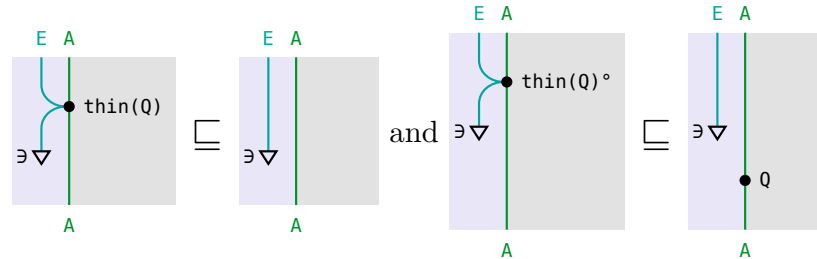
| the law                                                                                                                                                                                                                            | what it says                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| $X \in \frac{S}{\exists} \text{ thin}(Q) \iff X \in S$ and $S^\circ X \in Q^\circ \epsilon$                                                                                                                                        | everything kept is an $S$ -value, and every $S$ -value has a $Q$ -lower bound among the kept ones                     |
| $\text{thin}(Q) \exists \epsilon \exists$                                                                                                                                                                                          | everything a thinning keeps was in the set                                                                            |
| $\text{thin}(Q)^\circ \exists \epsilon \exists Q$                                                                                                                                                                                  | every element of the set has a $Q$ -lower bound among the kept ones                                                   |
| $Q \in R \implies \text{thin}(Q) \in \text{thin}(R)$                                                                                                                                                                               | the fewer pairs $Q$ relates, the fewer subsets count as thinnings                                                     |
| $1 \in Q \implies 1 \in \text{thin}(Q)$<br>$Q \in Q \implies \text{thin}(Q) \text{ thin}(Q) \in \text{thin}(Q)$                                                                                                                    | keeping everything is always a legal thinning                                                                         |
| $Q \in R \wedge 1 \in Q \wedge R \in R \implies \text{est}(R) = \text{thin}(Q) \text{ est}(R)$ $Q \in R, 1 \in Q, R \in R$ — (8.1.2a); weaker than the book's “both preorders”: $Q$ transitive is never used and $1 \in R$ follows | <b>thin-introduction</b> : thinning first cannot lose an $R$ -minimum                                                 |
| $\text{thin}(Q) \supseteq \text{est}(Q) \quad \frac{1}{\exists} \quad (8.2) \text{ — } (8.2)$                                                                                                                                      | <b>thin-elimination</b> : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on |
| $\frac{S}{\exists} \text{ thin}(Q) \supseteq \frac{S}{\exists} \text{ est}(R) \quad \frac{1}{\exists}$<br>(8.3), $R \cap (S^\circ S) \in Q$ — thin-elimination variant                                                             | the usable variant: $R$ need only refine $Q$ between values $S$ gives one argument                                    |
| $P(\text{thin}(Q)) \text{ union} \in \text{union thin}(Q) \quad (8.4)$                                                                                                                                                             | thinning each member set is a thinning of the union                                                                   |

#### §8.1.1. $X \in (S^\circ \exists) \text{ thin}(Q) \iff X \in S$ and $S^\circ X \in Q^\circ \epsilon$

(8.1.1a)



(8.1.1b)



(8.1.1a) at  $X \triangleq \text{thin}(Q)$ ,  $S \triangleq \exists$ : the left side is  $\text{thin}(Q) \in \text{thin}(Q)$ , since  $\exists \circ \exists = \wedge(\exists) = 1$

#### §8.1.2. $\text{est}(R) = \text{thin}(Q) \text{ est}(R)$ given $Q \in R, 1 \in Q, R \in R$

**$\text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R)$**  (8.1.2a)  
the  $\sqsubseteq$  half: keeping everything is a thinning —  $1 \sqsubseteq Q$

(a)  $1 \sqsubseteq \text{thin}(Q)$  — (8.1.1a)

**$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists$**  (8.1.2b)  
the  $\sqsubseteq$  half, first condition of the UP of  $\text{est}$  at  $X \triangleq \text{thin}(Q) \text{ est}(R)$  — est-up

(a)  $\text{est}(R) \sqsubseteq \exists$  — est-laws  
(b)  $\text{thin}(Q) \sqsubseteq \exists$  — (8.1.1a)

**$\in \text{thin}(Q) \text{ est}(R) \sqsubseteq R^\circ$**  (8.1.2c)  
the  $\sqsubseteq$  half, second condition —  $Q \sqsubseteq R$ ,  $R$  transitive

(a)  $\in \text{thin}(Q) \sqsubseteq Q^\circ \in$  — (8.1.1a)  
(b)  $\in \text{est}(R) \sqsubseteq R^\circ$  — est-laws  
(c)  $R$  transitive

**$\text{est}(Q) \frac{1}{\exists} \sqsubseteq \text{thin}(Q)$**  (8.2)  
the singleton holding a  $Q$ -least member of a set is a thinning of that set

**$\text{est}(Q) \frac{1}{\exists} \sqsubseteq \exists$**   
every member of the singleton is a member of the set

(a)  $\triangleq \text{thin}, \exists/\exists$  half  
(b)  $\frac{1}{\exists} \sqsubseteq 1$  — pow-laws;  $\text{est}(Q) \sqsubseteq \exists$  — est-laws

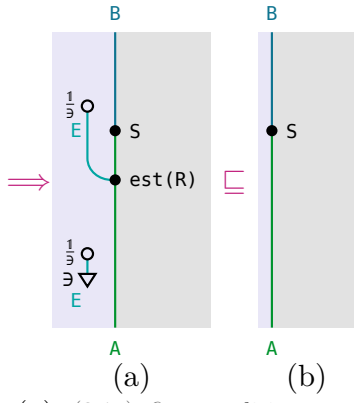
**$\in \text{est}(Q) \frac{1}{\exists} \sqsubseteq Q^\circ \in$**   
the singleton's member stands in  $Q$  to every member of the set

(a)  $\triangleq \text{thin}, \in \setminus (Q^\circ \in)$  half  
(b)  $\in \text{est}(Q) \sqsubseteq Q^\circ$  — est-up  
(c)  $\frac{1}{\exists} \sqsubseteq \in$  — pow-laws

$$\mathbf{RnS^\circ S \sqsubseteq Q \Rightarrow \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \sqsubseteq \frac{S}{\exists} \text{thin}(Q)} \quad (8.3)$$

given  $\mathbf{Rn(S^\circ S) \sqsubseteq Q}$ ,  $Q$  a preorder

$\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \exists \sqsubseteq S$   
every member of the singleton is a value  $S$  returns

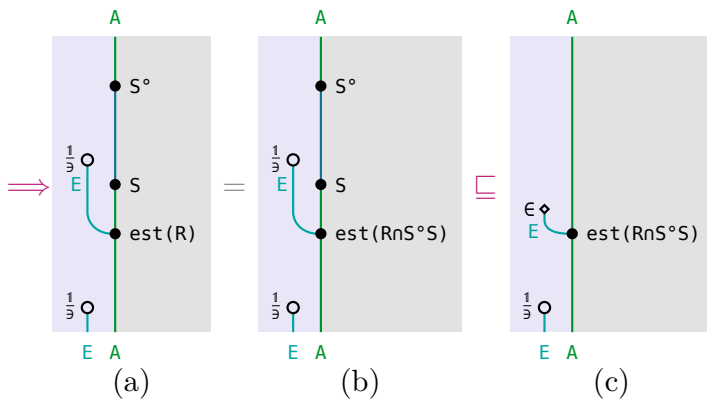


(a) (8.1a), first condition

(b)  $\frac{1}{\exists} \exists = 1$  — pow-laws;  $\frac{S}{\exists} \text{est}(R) \sqsubseteq S$  — est-laws

$$\mathbf{RnS^\circ S \sqsubseteq Q \Rightarrow S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \sqsubseteq Q^\circ \in}$$

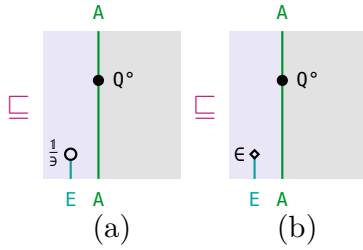
the singleton's member stands in  $Q$  to every value  $S$  returns at the same argument



(a) (8.1a), second condition

(b)  $\frac{S}{\exists} \text{est}(R) = \frac{S}{\exists} \text{est}(RnS^\circ S)$  — est-laws

(c)  $S^\circ \frac{S}{\exists} \sqsubseteq \in$  — pow-laws



(a)  $\in \text{est}(RnS^\circ S) \sqsubseteq (RnS^\circ S)^\circ$  — est-up;  $Rn(S^\circ S) \sqsubseteq Q$

(b)  $\frac{1}{\exists} \sqsubseteq \in$  — pow-laws

Theorem 8.1

$$QQ \sqsubseteq Q \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle$$

thinning at every step of the reduce is a thinning of the whole candidate set —  $S$  monotonic on  $Q$ ,  $Q$  a preorder

(8.1a) at  $X \triangleq \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle$ ,  $\langle (S) \rangle$  for its  $S$

(a) fusion, hylomorphism theorem  
 (b)  $S^\circ F(Q^\circ) \sqsubseteq Q^\circ S^\circ$  — mon-str, relator-laws  
 (c)  $S^\circ F(\epsilon) \sqsubseteq \epsilon$  — pow-laws  
 (d)  $\epsilon \text{thin}(Q) \sqsubseteq Q^\circ \epsilon$  —  $\triangleq \text{thin}$ , adj-all

Corollary 8.1

$$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge QQ \sqsubseteq Q \wedge RR \sqsubseteq R \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \text{est}(R) \sqsubseteq \langle \frac{(S)}{\exists} \text{est}(R) \rangle$$

the thinning fold refines the optimisation problem itself —  $S$  monotonic on  $Q$ ,  $Q \sqsubseteq R$ , both preorders

(a) thinning theorem  
 (b)  $\text{est}(R) = \text{thin}(Q) \text{est}(R)$  — (8.1a)

## §8.2. Paths in a layered network

(8.2a)

$F(A, X) = A + A \times X$ ,  $L = \text{list}^+$  with initial algebra  $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$ .  
 $\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle$ ,  $\text{consw}(a, (xs, n)) = (\text{cons}(a, xs), \text{wt}(a, \text{head}(xs)) + n)$ .  
 $\text{cost} \triangleq \langle [\text{wrapz}, \text{consw}] \rangle \pi_2$ ,  $\langle [\text{wrapz}, \text{consw}] \rangle = \langle 1, \text{cost} \rangle$ ,  $R \triangleq \text{cost} \leq \text{cost}^\circ$ .  
 $Q \triangleq Rn(\text{head head}^\circ)$ ,  $S \triangleq F(1, \exists)\alpha$ ,  $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$ ,  $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$ ,  $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{est}(R)$ .

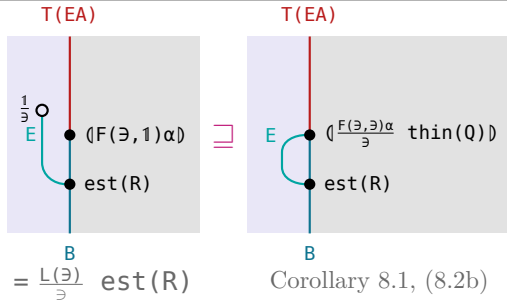
(8.2b)

| the law                                                                         | what it says                                                                                                                                                        |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$                       | $F(\exists, 1) \alpha$ is monotonic on $Q$ ; on $R$ it is not, since the next edge can cost arbitrarily much                                                        |
| $S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$ | $S \text{ head}$ is simple, which gives $Rn(S^\circ S) \sqsubseteq Q$ : between two paths $S$ builds from one argument, equal cost and equal head already means $Q$ |

$$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge QQ \sqsubseteq Q \wedge R^\circ R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(F(\exists, 1)\alpha, Q) \wedge Rn(F(1, \exists)\alpha)^\circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \downarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \quad (8.2c)$$

$$\text{est}(R) \sqsubseteq \text{est}(R) \sqsubseteq \frac{F(\exists, 1)\alpha}{\exists} \text{est}(R)$$

a least-cost path in a layered network, as a fold over the layers



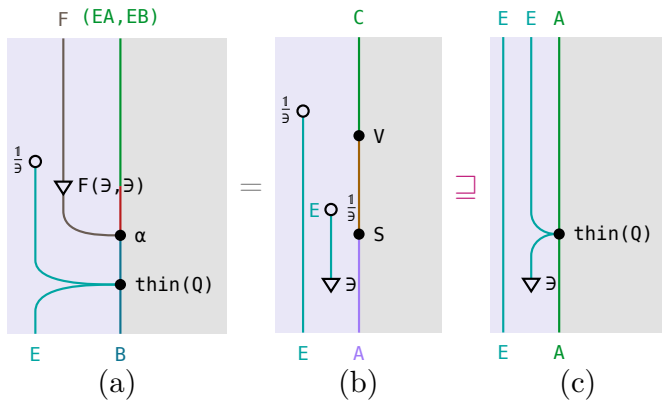
$\sqsubseteq$  (8.2d) under  $\downarrow$   $\sqsupset$  monotonic: the whole chain runs inside the reduce, and the  $\text{est}(R)$  behind it never moves.

$$Rn(F(1, \exists)\alpha)^\circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{est}(R) \sqsubseteq \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q) \quad (8.2c)$$

thinning the algebra of a layered network costs no more than taking the program's two cases

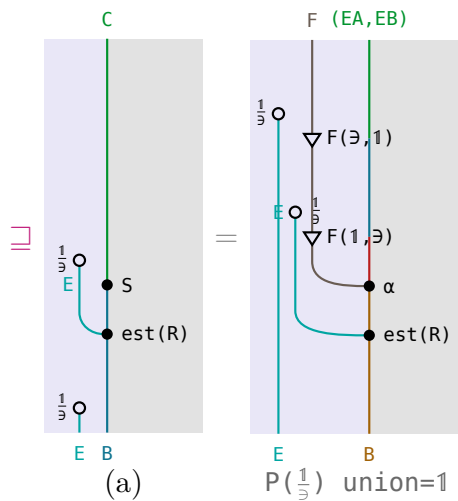
$$\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \text{union} \sqsubseteq \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q)$$

thinning each set the algebra's parts build, then taking the union, is below thinning the one set the whole algebra builds



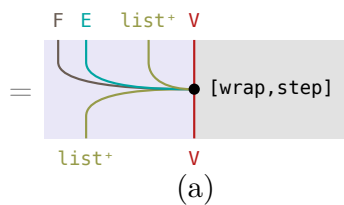
$$Rn(F(1, \exists)\alpha)^\circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{est}(R) \sqsubseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \text{union}$$

choosing an R-least member of each of those sets is below thinning each and taking the union



$$\text{cp } P(\alpha) \text{est}(\text{path}) = [\text{wrap}, \text{step}]$$

at the layered network, choosing an R-least path from each case is the program's two cases  $[\text{wrap}, \text{step}]$



$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} \triangleq \text{cost}$$

## §8.3. Implementing thin

(8.3a)

$\text{setify} : [A] \rightarrow EA$ ,  $\text{cup} : EA \times EA \rightarrow EA$ ,  $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$ ,  $\text{listcp} : F(L) \rightarrow LF$ ,  $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P$  for  $P$  a connected preorder.

$\text{thinlist}(Q)$  is any  $\text{thinlist}(Q) \sqsubseteq \text{subseq}$  with  $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$ ; one is  $\llbracket \text{nil}, \text{bump } Q \rrbracket$ ,  $\text{bump } Q (a, []) = [a]$ ,  $\text{bump } Q (a, [b] \# xs) = (b \sqsubseteq_Q a \rightarrow [a] \# xs, a \sqsubseteq_Q b \rightarrow [b] \# xs, [a] \# [b] \# xs)$ .

**Binary thinning** data:  $S = (f_1 p_1) \cup (f_2 p_2)$  with  $p_1, p_2$  coreflexive;  $Q$  a preorder with  $Q \sqsubseteq R$  and both  $f_1 p_1, f_2 p_2$  monotonic on  $Q$ ;  $P$  a connected preorder with both  $f_1, f_2$  monotonic on  $P$ ;  $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$ .

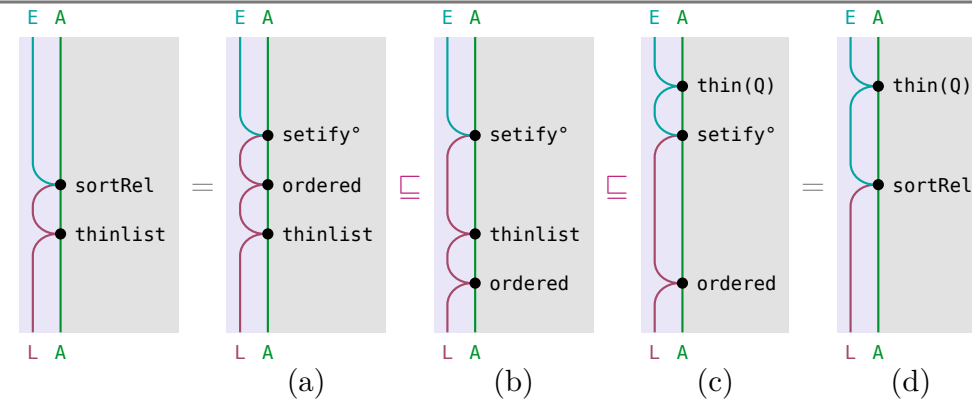
(8.3b)

| the law                                                                                                  | what it says                                                                                                       |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| $\text{thinlist}(Q) \text{ xs} = [\text{minlist}(Q) \text{ xs}]$<br>(8.5), $Q$ connected, $xs$ non-empty | what thinning should come to when it can: one element                                                              |
| $\text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P)$ (8.6) — (8.6)            | thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys |
| $\text{sort}(P) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)                                      | a minimum of the sorted list is a minimum of the set                                                               |
| $\text{sort}(f P f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(P)$ (8.8)                        | shunt a function through a sort                                                                                    |
| $\text{sort}(P) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(P)$ (8.9), $p$ coreflexive               | filtering a sorted list sorts the restricted set                                                                   |
| $(\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \sqsubseteq \text{cup sort}(P)$ (8.10)          | merging two sorted lists sorts their union                                                                         |
| $F(\text{sort}(P)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(FP)$<br>(8.11), $F$ linear       | $\text{listcp}$ is the list implementation of the cartesian product $\text{cp}(F)$                                 |

**Map(setify)  $\wedge$  Coreflexive(ordered)  $\wedge$  thinlist  $\sqsubseteq \text{subseq} \wedge$  ordered subseq  $\sqsubseteq \text{subseq ordered} \wedge$  thinlist** (8.6)

**setify  $\sqsubseteq \text{setify thin}(Q) \Rightarrow \text{sortRel thinlist} \sqsubseteq \text{thin}(Q) \text{ sortRel}$**

a thinning of the sorted list lists a thinning of the set —  $P$  a connected preorder,  $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ .

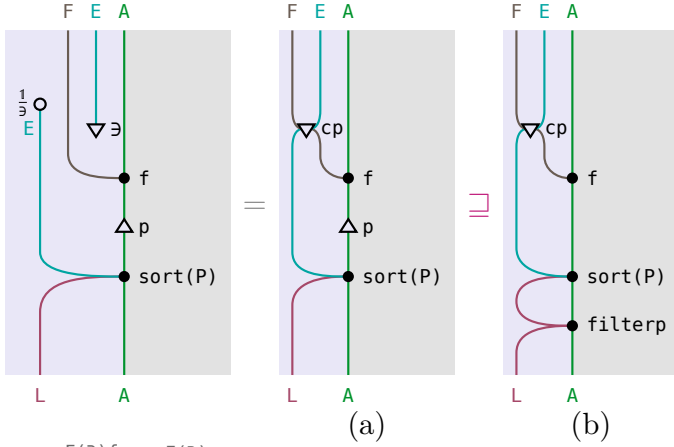


- (a)  $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ — } \triangleq \text{thinlist}$   
 (b)  $\text{ordered } P \text{ thinlist}(Q) \sqsubseteq \text{thinlist}(Q) \text{ ordered } P \text{ — } \triangleq \text{thinlist}$   
 (c)  $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q) \text{ — } \triangleq \text{thinlist}$ , dom-laws, triple-chains  
 (d)  $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ — } \triangleq \text{thinlist}$

Lemma 8.1

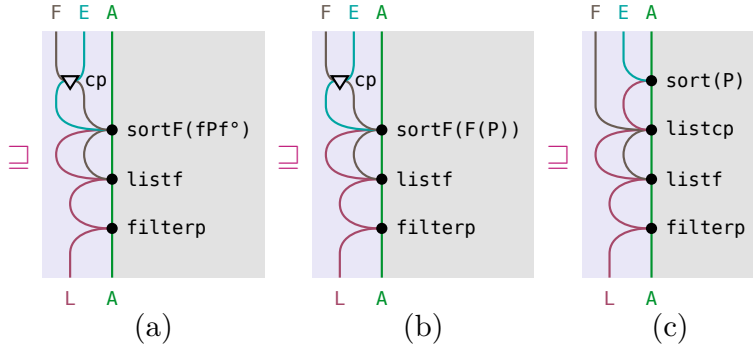
$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge$   
 $\text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge \text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \wedge F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}$   
 $\text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$   
 one sorted list built from sorted arguments, instead of a set built and then sorted —  $f : FA \rightarrow A$  monotonic  
 on  $P$ ,  $p$  coreflexive,  $F$  linear.

$\text{Map}(f) \wedge \text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \Rightarrow \text{cp } P(f) \text{sort}(P) \text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$   
 sorting the  $f$ -images of the product and filtering by  $p$  is below sorting the set  $F(\exists)fp$  builds



- (a)  $\frac{F(\exists)fp}{\exists} = \frac{F(\exists)}{\exists} E(fp)$  — pow-laws,  $\triangleq$  thinlist  
 $E(f) = P(f)$ ,  $f$  a map — powrel-laws  
 (b)  $\text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P)$  — (8.3b)

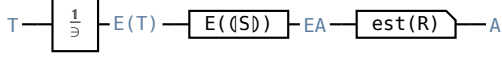
$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge \text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge$   
 $F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp} \text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \text{cp } P(f) \text{sort}(P) \text{filterp}$   
 combining the arguments' sorted lists by  $\text{listcp}$ , mapping  $f$  and filtering is below sorting the  $f$ -images and filtering



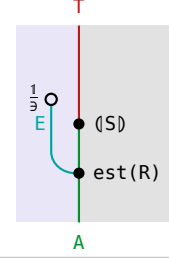
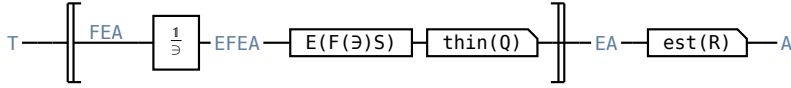
- (a)  $\text{sort}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P)$  — (8.3b)  
 (b)  $FP \sqsubseteq fP f^\circ$  — mon-str,  $\triangleq$  thinlist  
 (c)  $F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sort}(FP)$  — (8.3b)

$\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R \circ R \circ \sqsubseteq R \circ \wedge \text{MonoAlg}(f_1 \ p_1, Q) \wedge \text{MonoAlg}(f_2 \ p_2, Q)$  Theorem 8.2  
 $\wedge (\forall X. \forall Y. X \sqsubseteq Y \implies \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \text{sortF}(f_1 \ P f_1^\circ) \text{listf}_1 \sqsubseteq P(f_1) \text{sort}(P) \wedge \text{sortF}(f_2 \ P f_2^\circ) \text{listf}_2 \sqsubseteq P(f_2) \text{sort}(P) \wedge$   
 $\text{sort}(P) \text{filterp}_1 \sqsubseteq E(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filterp}_2 \sqsubseteq E(p_2) \text{sort}(P) \wedge F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}$   
 $\text{sortF}(F(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup } \text{sort}(P) \wedge$   
 $\text{sort}(P) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(P) \wedge \text{sort}(P) \text{minlist}(R) \sqsubseteq \text{est}(R) \wedge g_1 = \text{listf}_1 \text{ filterp}_1 \wedge$   
 $g_2 = \text{listf}_2 \text{ filterp}_2 \wedge S = f_1 \ p_1 \cup f_2 \ p_2 \implies (\text{listcp}(g_1, g_2) \text{merge}(P) \text{thinlist}(Q))$   
 $\text{minlist}(R) \sqsubseteq \frac{(S)}{S} \text{est}(R)$   
 a fold on sorted lists of partial solutions, thinned at every step — at  $\triangleq \text{thinlist}$ 's binary thinning data.

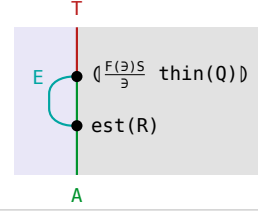
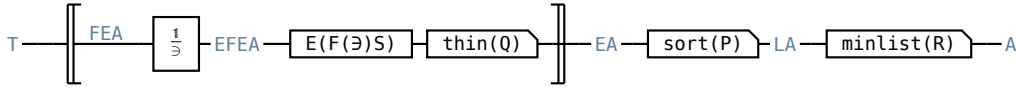
## circuit



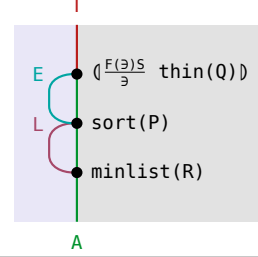
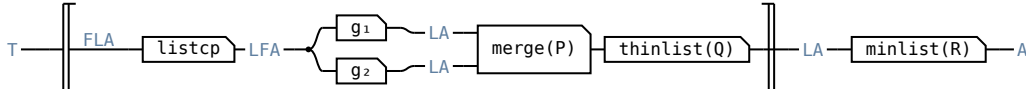
## Hinze–Marsden


 $\sqsubseteq$ 


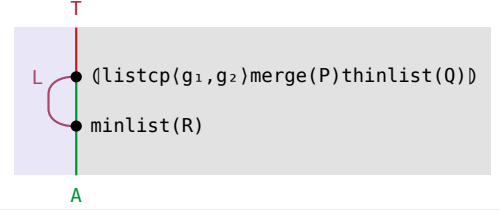
Corollary 8.1 at  $f_1 p_1$  and  $f_2 p_2$  monotonic on  $Q \triangleq \text{thinlist}$


 $\sqsubseteq$ 


$\text{sort}(P) \text{minlist}(R) \sqsubseteq \text{est}(R) \text{ — (8.3b) at its } Q \triangleq R$


 $\sqsubseteq$ 


fusion at (8.3c)



## §8.4. The knapsack problem

10

(8.4b)

| the law                                                                                                                      | what it says                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| $(1 \times R) \text{ (cons (within } w)) \sqsubseteq \text{cons (within } w)R$<br>FALSE                                      | not monotonic on R: a selection of greater value need not still fit once one more item goes in |
| $(1 \times Q) \text{ (cons (within } w)) \sqsubseteq \text{cons (within } w)Q$<br>$(1 \times Q) \pi_2 \sqsubseteq \pi_2 Q$ , | both halves are monotonic on Q once ties in value are broken by weight                         |

$$\begin{aligned} & 0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \wedge (\forall X. \forall Y. X \subseteq Y \Rightarrow \text{sortF}(X) \subseteq \text{sortF}(Y)) \wedge \\ & \text{sortF}([\text{nil}, \text{cons}]R[\text{nil}, \text{cons}]^\circ) \text{listf}_1 \subseteq P([\text{nil}, \text{cons}]) \text{sortP} \wedge \\ & \text{sortF}([\text{nil}, \pi_2]R[\text{nil}, \pi_2]^\circ) \text{listf}_2 \subseteq P([\text{nil}, \pi_2]) \text{sortP} \wedge \text{sortP} \text{filterp}_1 \subseteq E(\text{within}) \text{sortP} \wedge \\ & F(\text{sortP}) \text{listcp} \subseteq \text{cp} \text{sortF}(F(R)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \subseteq \text{cup} \text{sortP} \wedge \text{sortP} \\ & \text{thinlist} \subseteq \text{thin}(Q(\text{vol}, \text{wt})) \text{sortP} \wedge \text{sortP} \text{minlist} \subseteq \text{est}(R) \Rightarrow (\text{listcp}(\text{listf}_1 \text{filterp}_1, \text{listf}_2) \text{mergeP} \\ & \text{thinlist}) \text{minlist} \subseteq \frac{\text{subseq} \text{ within}}{\exists} \text{est}(R) \end{aligned}$$

the knapsack problem, as a fold that thins the packings kept at each item

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Hinze–Marsden |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $  \begin{array}{c}  \boxed{\text{[Item]}} \text{---} \boxed{\frac{1}{\exists}} \text{---} \boxed{E(\text{subseq within})} \text{---} \boxed{E[\text{Item}]} \text{---} \boxed{\text{est}(R)} \text{---} \boxed{\text{[Item]}} \\  \text{subseq (within w)} \\  \exists \quad \text{est}(R)  \end{array}  $                                                                                                                                                                                                                                                                                                                                                                                                               |               |
| $  \begin{array}{c}  \boxed{\text{[Item]}} \text{---} \boxed{\frac{1}{\exists}} \text{---} \boxed{E(\langle [\text{nil}, \text{cons}] \text{ within } u \text{ } [\text{nil}, \pi_2] \rangle)} \text{---} \boxed{E[\text{Item}]} \text{---} \boxed{\text{est}(R)} \text{---} \boxed{\text{[Item]}} \\  = \quad \langle [\text{nil}, \text{cons}] \text{ (within w)} \cup [\text{nil}, \pi_2] \rangle \\  \exists \quad \text{est}(R) \\  \text{fusion, weights non-negative.}  \end{array}  $                                                                                                                                                                                                                             |               |
| $  \begin{array}{c}  \boxed{\text{[Item]}} \text{---} \boxed{\frac{1}{\exists}} \text{---} \boxed{\text{listcp}} \text{---} \boxed{\text{list}(F[\text{Item}])} \text{---} \boxed{\text{listf}_1} \text{---} \boxed{\text{filterp}_1} \text{---} \boxed{\text{list}[\text{Item}]} \text{---} \boxed{\text{mergeP}} \text{---} \boxed{\text{thinlist}} \text{---} \boxed{\text{list}[\text{Item}]} \text{---} \boxed{\text{minlist}} \text{---} \boxed{\text{[Item]}} \\  \sqsubseteq \quad \langle \text{listcp} \langle g_1, g_2 \rangle \text{ merge } R \text{ thinlist}(Q) \rangle \text{ minlist}(R) \\  \text{binary thinning theorem, at } P \triangle R, F \text{ linear, } Q \text{ from (8.4b)}  \end{array}  $ |               |
| $  \begin{array}{c}  \langle [\text{nil}, \text{cpr} \langle h_1, h_2 \rangle \text{ merge } R \text{ thinlist}(Q)] \rangle \text{ minlist}(R) \\  = \quad \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{nil}), h_i] \text{ --- } \triangle \text{ within; minlist}(R) \\  \text{is head, packings coming out in descending value}  \end{array}  $                                                                                                                                                                                                                                                                                                                                                   |               |

## §8.5. The paragraph problem

(8.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.
new(a,xs)=[a]#xs, glue(a,xs)=[a]#head(xs)#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

(8.5b)

| the law                                                                                                                                                                                                              | what it says                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| $(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$                                                                                                                                            | glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this |
| $(1 \times Q) \text{ new} \sqsubseteq \text{new } Q$<br>$(1 \times Q) (\text{glue } (ok \ w)) \sqsubseteq \text{glue } (ok \ w) Q$<br>$\text{cons monotonic on } \text{collect} \sqsubseteq \text{collect}^\circ, .$ | both halves are monotonic on Q once ties in waste are broken by the first line                                           |
| merge $T = \text{cat}$ ; $P \triangleq \text{head prefix head}^\circ$ also serves                                                                                                                                    | $T$ needs no sorting at all, and $\text{prefix}$ is a linear order on first lines of paragraphs of one input             |

(8.5c)

$(\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap}$   
 $\text{wrap,new}]T[\text{wrap wrap,new}]^\circ) \text{listf}_1 \sqsubseteq P([\text{wrap wrap,new}]) \text{sortP} \wedge \text{sortF}([\text{wrap wrap,glue}]T[\text{wrap}$   
 $\text{wrap,glue}]^\circ) \text{listf}_2 \sqsubseteq P([\text{wrap wrap,glue}]) \text{sortP} \wedge \text{sortP filterp}_2 \sqsubseteq E(ok) \text{sortP} \wedge F(\text{sortP}) \text{listcp} \sqsubseteq \text{cpr}$   
 $\text{sortF}(F(T)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq \text{cup sortP} \wedge \text{sortP thinlist} \sqsubseteq \text{thin}(Q(\text{len},w)) \text{sortP} \wedge \text{sortP}$   
 $\text{minlist} \sqsubseteq \text{est}(R) \Rightarrow ([\text{listcp}(\text{listf}_1, \text{listf}_2 \text{ filterp}_2) \text{ mergeP thinlist}] \text{minlist} \sqsubseteq \frac{\text{partition fits}}{3} \text{est}(R))$   
 a paragraph laid out as a fold that thins the layouts kept at each word

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Hinze–Marsden |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{3}} E(\text{list}^*(\text{Word})) \xrightarrow{E(\text{partition fits})} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $\frac{\text{partition list}^*(\text{fits } w)}{3} \text{est}(R)$                                                                                                                                                                                      |               |
| $\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{3}} E(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap,new} u [\text{wrap wrap,glue}ok])} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $= \frac{([\text{wrap wrap,new} u (\text{glue } (ok \ w))])}{3} \text{est}(R)$ <p>fusion, every word fits on a line by itself.</p>                                                                                         |               |
| $\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{3}} E(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap,new} u [\text{wrap wrap,glue}ok])} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $= \frac{([\text{wrap wrap,new} u ([\text{wrap wrap,glue} (ok \ w))])}{3} \text{est}(R)$ <p>the algebra as <math>(f_1 p_1) \cup (f_2 p_2)</math>, <math>p_1 \triangleq 1 \text{ — } \triangleq \text{thinlist}</math>.</p> |               |
| $\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{3}} E(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap,new} u [\text{wrap wrap,glue}ok])} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $\sqsubseteq ([\text{listcp } (g_1, g_2) \text{ cat thinlist}(Q)] \text{minlist}(R))$ <p>binary thinning theorem, at <math>P \triangleq T</math> with merge <math>T = \text{cat}</math>, <math>Q</math> from (8.5b)</p>    |               |
| $([\text{start, cpr } (h_1, h_2) \text{ cat thinlist}(Q)])$ $= \text{minlist}(R)$ <p><math>\text{listcp} = \text{wrap} + \text{cpr}</math>, <math>g_i</math> along the coproduct — <math>\triangleq \text{partition}</math></p>                                                                                                                                                                                                                                                                       |               |

